CEC 320 Course Info for Spring 2026

1 General Info

- Class meeting time:
  - Sec 1: MWF 9:00 to 9:50 am
  - Sec 2: MWF 10:00 to 10:50 am
- Room: CoB 124
- Instructor: Dr. Jianhua Liu, Associate Professor of Electrical and Computer Engineering
- Instructor's office: LB 349
- Instructor's email: jianhua.liu@erau.edu or liu620@erau.edu
- Instructor's phone number: 226-7713 (Work)
- Instructor's office hours: MWF 2:30 pm to 4:50 pm; appointment on Zoom.
- TAs' office hours: TBD

2 Prerequisites and Corequisites

- Prerequisite: CEC 220
- Corequisite: CEC 322

3 Catalog Course Description

Microprocessor or microcontroller organizations and programming. Members of a current microprocessor or microcontroller family. Assembly language programming techniques for microprocessors or microcontrollers. General purpose input and output, serial communications, and other peripherals. Hardware and software relationships.

4 Learning Outcomes

After taking this course, students will be able to do the following:

- Describe the microprocessor or microcontroller architecture and related components.
- Apply the programmer's model of a microprocessor or microcontroller.
- Map the C programming language to the instruction set of a microprocessor or microcontroller.
- Program assembly language functions using the instruction set.
- Implement programs dealing with interrupts.
- Operate general-purpose input output (GPIO) ports.
- Use serial communications, timers, and pulse width modulations.

5 Required Textbook and Class notes

- A collection of class notes in a single PDF file, called **the Collection**, is provided.

  - Notes are organized according to lecture—each lecture has a corresponding chapter.
  - Used for readings to get full information/understanding of the contents of lectures. Reading of each chapter is highly encouraged, especially for those interested in a fully understanding the topics.

- A compilation of additional notes in a single PDF file, called **the Companion**, is also provided.
  - Notes of DevTools are provided here.
  - No need to read everything; for those with extra time and interest, there are a lot of additional information.

These two PDF files may contain too much information for a single semester. To help students master the fundamentals, we also provide lecture notes, which are most essential topics extracted from *the Collection*. These notes are provided in two versions. The *before-class-handout* (BCH) version, which is only partially populated, can be used to help students to reduce the load of taking notes in class. When the problems are worked out in the class, we will have the *post-lecture-workout* (PLW) version. Note that the traditional way to taking notes, which copies the white-board writings may be too time consuming for this class as we have a lot code snippets and diagrams.

6 Reference Books

- Yifeng Zhu, *Embedded Systems with ARM Cortex-M Microcontrollers in Assembly Language and C*, Third Edition, E-Man Press LLC, 2017. This was the textbook we used before with a comprehensive discussion of many topics. A good read if you plan to have a career in this aera.
- ARM, *Cortex-M4 Devices—Generic User Guide*, 2011. Download available at https://developer.arm.com/documentation/dui0553/b. Chapter 3, The Cortex-M4 Instruction Set, is of the most interests to us. It lists all the assembly instructions of the ARM microprocessor we use.
- Joseph Yiu, *The Definitive Guide to ARM Cortex-M3 and Cortex-M4 Processors*, Third Edition, Newnes, 2014. This is really the definitive guide for the topic. However, this book is not an easy read before finishing this class.
- John Sonmez, *The Complete Software Developer's Career Guide: How to Learn Programming Languages Quickly, Ace Your Programming Interview, and Land Your Software Developer Dream Job*, 2017. This is a highly recommend reading which contains a lot of great suggestions for having a successful career for not not software developers but in other areas as well.

7 Required Hardware

- A laptop computer running Windows, macOS, or Linux.
- An MCU board and related hardware can be borrowed from the TAs. If you are interested in purchasing the MCU board, NUCLEO-G431KB, there is the link: https://www.st.com/en/evaluation-tools/nucleo-g431kb.html#sample-buy.

8 Learning of the Contents

University education is not the same as job training in companies or trade schools. As such, we have an additional goal with this class—the practice of *life-lone learning*, which is especially important for topics like

microprocessor architecture and programming.

The real-world topics of microprocessor and programming cover a vast scope, and there is no way to learn everything in a single class. What we teach here are *fundamentals*, with which you can move on to teaching yourself whatever topics you are interested in. A good start is reading the book by Joseph Yiu, listed above.

Different people has different preferences for learning. Yet, reading is considered by many as the most important skill for life-long learning.

The instructor has spent a significant amount time refining the class notes to two PDF files, which not only cover the basics but a lot of important application tricks. If you are well-prepared with academic reading, you should have no issues understanding the contents. If you are less prepared, taking this class can be a great chance to improve your reading. There are some suggestions regarding reading in *the Compilation*.

There are two learning approaches for this class:

- Reading-based learning. For this approach, you need to read the BCH version carefully and try to get the ideas by reading *the Collection*. When you have issues, note them down to get them addressed in class. To better use the class time, you can use it to do your homework problems. You can also conditionally skip some lectures if you use this approach, as described below.
- Lecture-based learning. Use this approach only if you have trouble reading. Yet, you still need to try your best to read as much as possible, as one of the goals of taking this class is to practice life-long learning by reading.

We have many small topics in this class that look similar but are different. Regardless of the approach you use, knowing the definitions of everything is very important. You are encouraged to make mind maps or cheatsheets so that you can summarize, compare, and contrast these topics.

9 Class Meeting Pattern

Class meeting (CM) time is organized in units; each unit has two lecture meetings (LMs) and a practice meeting (PM).

10 Assessment of Learning Outcomes

10.1 General guideline

- There will be in-class activities during both LMs and PMs to help students learn well in class.
- There will be unit-based homework assignments.
- There will be programming project assignments which have very low weighting and are mostly used for practice purposes.
- There will be a quiz every three units except the first one, which first one which covers two units.
  - There will be no special exercises or makeups as the in-class activities in practice meeting and HW problems are designed for practices already.
- There will be a tentatively mandatory final exam.

  - If you do not do well, this will not affect your class grade.
  - Location is the classroom, and time is prescribed at https://myerauedu.sharepoint.com/sites/Registrar/SitePages/daytona-term-final-exam-schedules.aspx:
    * 4/27/26, Monday: 8:00 am - 10:00, Sec 1 (MWF 9 to 9:50 am)
    * 4/29/26, Wednesday: 8:00 am - 10:00, Sec 2 (MWF 10 to 10:50 am)

10.2 In-class activities

- In-class activities include (1) Unannounced attendance check, (2) Unannounced quizzes, and (3) Participation in solving class problems.
- Attendance will be 80%, and the other 20% will be participation based.

10.3 Practice meeting activities

- PM can be used as a cushion of teaching.
- We will answer questions regarding the HW problems.
- We will focus on solving ONE new problem (not in HW assignment) in a systematic way.
  - Student will do the problem.
  - Answer will be provided.
  - In-class peer grading is performed. If wrong, both the problem solver and the grader will lose points. Points:
    * Attendance: 3
    * Correct formula: 1
    * Correct answer: 1
  - Paper-based or electronic submission (scan to PDF) is needed.

10.4 Homework assignments

Homework problems will usually cover the contents of the most *recent two* lectures in the unit. For example:

- HW 1 covers contents of Lectures 2 and 3.
- HW 2 covers contents of Lectures 4 and 5.

HW assignments will not take up too much time if you truly understand the topics.

You are encouraged to discuss HW problems with your peers, but you cannot copy solutions from them. You have to make sure you understand everything you write on the answer sheet.

You can use LLM to help you out; yet, you should fully understand the solutions.

10.5 Class projects

Like homework assignments, these assignments do not take much time if you have a good background with programming. If you struggle, it does not mean you are bad with programming; it just means that you are less-prepared. To be better prepared, you need to practice and get help.

You will have two weeks to finish the programming projects. We will not discuss the projects in class. Yet, the instructor will collect questions regarding the project at the 5th day of its assignment and then make a video to address these questions at the 7th day. So, please get started ASAP and post your questions.

You can use LLM to help you out; yet, you should fully understand the code and explain your solutions.

10.6 Quizzes

Quizzes are based on the HWs practiced in the previous units.

- No notes, computer, or smart phone can be used. NO LLMs.
- Formula sheets can be used. For Quiz n, you can use n pieces of Letter-sized papers for formula sheets, which can be double-sided.

10.7 Final exam

The final exam is tentatively mandatory as we may have to collect ABET artifacts.

- Formula sheet:
  - No notes, computer, or smart phone can be used.
  - You can use 7 pieces of letter-sized paper for formula sheet, which can be double-sided.
- Coverage:
  - The difficulty level will be similar to those of the unit-based quizzes.
  - Problems may include everything covered in the entire semester, including HW problems not covered in quizzes.
  - The amount work will be about 2.5 times of a normal unit-based quizzes. Quizzes are designed for 40 minutes and the final for 120 minutes.

11 Class Schedule

Below, we only list the Lecture number in the course schedule. For details of contents of each lecture, see the content map at the end of this document.

Date and Day	Lecture	Others
1/7/26, Wednesday	CM 1: LM1, Ch 1. Intro to class	CM = Class meeting
1/9/26, Friday	CM 2: LM2, Ch 2. UART	LM = Lecture meeting
1/12/26, Monday	CM 3: LM3, Ch 3. MCU architecture	
1/14/26, Wednesday	CM 4: PM 1	PM = Practice meeting
1/16/26, Friday	CM 5: LM4, Ch 4. Data rep	
1/19/26, Monday	No class	Martin Luther King Jr. Day
1/21/26, Wednesday	CM 6: LM5, Ch 5. Ptrs & Unit test	HW 1 due

Date and Day	Lecture	Others
1/23/26, Friday	CM 7: PM 2	
1/26/26, Monday	CM 8: LM 6, Ch 6. GPIO	
1/28/26, Wednesday	CM 9: LM 7, Ch 7. Bitwise ops	HW 2 due
1/30/26, Friday	CM 10: PM 3	
2/2/26, Monday	CM 11: QZ 1	QZ 1 covers basics, HWs 1 & 2
2/4/26, Wednesday	CM 12: LM 8, Ch 8. Interrupt	
2/6/26, Friday	CM 13: LM 9, Ch 9. Preemption	HW 3 due
2/9/26, Monday	CM 14: PM 4	
2/11/26, Wednesday	CM 15: LM 10, Ch 13a Real numbers	
2/13/26, Friday	CM 16: LM 11, Ch 13b Real numbers	HW 4 due
2/16/26, Monday	No class	President's Day
2/18/26, Wednesday	CM 17: PM 5	
2/20/26, Friday	CM 18: LM 12, Ch 14. C/ASM prog	
2/23/26, Monday	CM 19: LM 13, Ch 15. Arithmetic	HW 5 due
2/25/26, Wednesday	CM 20: PM 6	
2/27/26, Friday	CM 21: QZ 2	Qz 2 covers HWs 3, 4, & 5
3/2/26, Monday	CM 22: LM 14, Ch 16. Shift & Op2	
3/4/26, Wednesday	CM 23: LM 15, Ch 17. Bitwise logic	HW 6 due
3/6/26, Friday	CM 24: PM 7	
3/9/26, Monday	CM 25: LM 16, Ch 18. Status flags	
3/11/26, Wednesday	CM 26: LM 17, Ch 19. Condition test	HW 7 due
3/13/26, Friday	CM 27: PM 8 (online)	
3/16/26, Monday	No class	Spring Break
3/18/26, Wednesday	No class	Spring Break
3/20/26, Friday	No class	Spring Break
3/23/26, Monday	CM 28: LM 18, Ch 20. LDR/STR 1	
3/25/26, Wednesday	CM 29: LM 19, Ch 21. LDR/STR 2	HW 8 due
3/27/26, Friday	CM 30: PM 9	
3/30/26, Monday	CM 31: QZ 3	QZ 3 covers HWs 6, 7 and 8
4/1/26, Wednesday	CM 32: LM 20, Ch 22. C ++/– ops	
4/3/26, Friday	CM 33: LM 21, Ch 23. ASM call fns	HW 9 due

Date and Day	Lecture	Others
4/6/26, Monday	CM 34: PM 10	
4/8/26, Wednesday	CM 35: LM 22, Ch 24. Branching	
4/10/26, Friday	CM 36: LM 23, Ch 25. Selection	HW 10 due
4/13/26, Monday	CM 37: PM 11	
4/15/26, Wednesday	CM 38: LM 24, Ch 26. Loop control	
4/17/26, Friday	CM 39: LM 25, Ch 27. Flow control	HW 11 due
4/20/26, Monday	CM 40: PM 12	
4/22/26, Wednesday	CM 41: QZ 4	QZ 4 covers HWs 9, 10 and 11
4/23/26, Thursday	No class	HW 12 due (online)

HW soln will be posted online the next day of submission. If you don't see it, please send the instructor a reminder email.

12 Grading Weighting and Letter Grade Assignment

Your class grade is the weighted average of various assessments, as shown below:

Assessment	Points	Assessment	Points
PM	10 points	HW	10 points
Quizzes	40 points	Final exam	40 points
Class projects	2 points		

- Total points: 102
- Grading Scale: 90+ => A, 80+ => B, 70+ => C, 60+ => D, 60- => F.

13 Class Policies

13.1 Meeting time policies

- In the class, no electronic distractions, such as texting or web browsing, will be allowed.
- Small talks that are not related to the subject of the practices is not allowed in the classroom. Repeated violations can result in the dropping of the student from the class.
- Students are encouraged to ask questions when they have.
  - If the questions are only specific to very few students, these students are invited to visit the instructor during the office hours.

13.2 Missing assignments

- No makeups for missing HW and class projects as they can be submitted online.
- Everyone can only make up ONE missing quiz for the entire semester.

13.3 Submission of work

For all the submissions, students are expected to act in a professional manner, including:

- Submitting work on time. Assignments will be given enough time to finish. No late submission is accepted.
- Active, collaborative learning is proven effective for improved learning. Students are encouraged to form teams of 2 to 3 students to discuss ideas of homework problems and projects in and out of class. Yet, each student should write their solutions independently.
- All work turned in for grading must be done neatly and professionally on Letter-sized paper, leaving 1-inch margins for making copy and binding. For electronic submission, please submit PDF only except otherwise specified. As a practice for AI-assisted teaching, please follow the guidelines below:
  - General requirements:
    * Should be LLM friendly (via scanning for paper-based submission).
    * Use appropriate size and spacing to write solution on the provided/indicated space.
  - For problem solving type:
    * Numerate the used formula (eqn/EQN) with clearly labeled variables if applicable.
    * Numerate the labeled final solution answer (ans/ANS).
  - For programing type:
    * Typing program code neatly following the 1TBS (one true brace style) format (with the 4-space indentation), as illustrated below.
    * Use correct syntax.

```
int abs(int x) {
    if (x >= 0) {
        return x;
    } else {
        return -x;
    }
}
```

13.4 Academic Integrity

ERAU is committed to intellectual integrity and considers academic dishonesty a very serious offense. Such offenses include cheating (accepting or giving unauthorized assistance in preparing assignments), or plagiarism (taking the ideas, writings, and/or work of another and representing them as one's own without appropriate acknowledgment). A student who cheats or commits any form of academic fraud will be subject to a failing grade for the course. In addition, the incident will be reported to the Dean of Students. If any other academic integrity violations have been documented, the student will be recommended for dismissal.

14 Coverage of the Collection

Figure 1: Content map of CEC 320 lectures.